

Project 3 : Extending Deep Generative Markov State Models: Non-Markovian and Continuous Dynamics

Supervisor - Grégoire SERGEANT-PERTHUIS

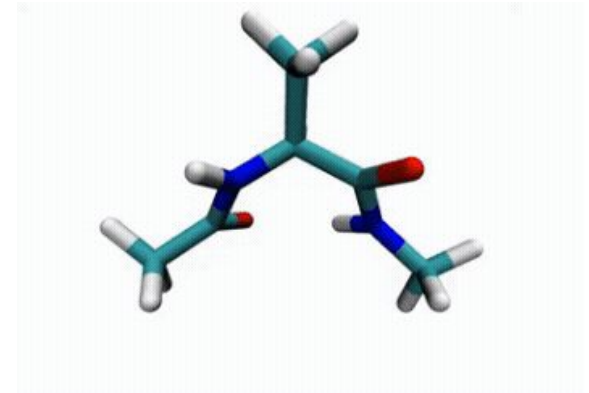
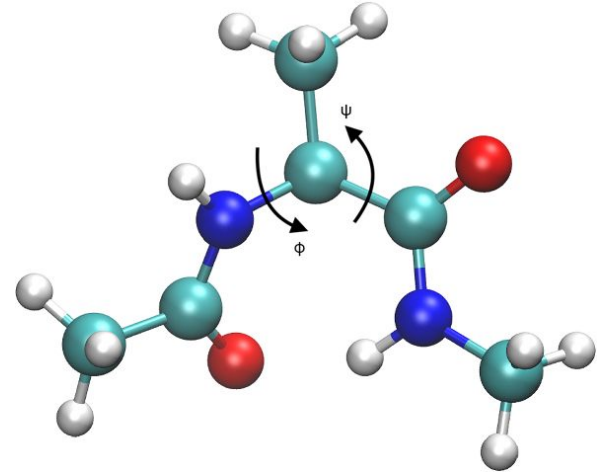
GABILLY Marie
LECENE Maël

Sorbonne University
25-26



Introduction - trajectory of alanine dipeptide

- Alanine dipeptide
 - 2 dihedral angles
-
- Succession states : trajectory
 - Wobbling -small changes between states-



Introduction - Context

Markov state models

high dimensional MD trajectories

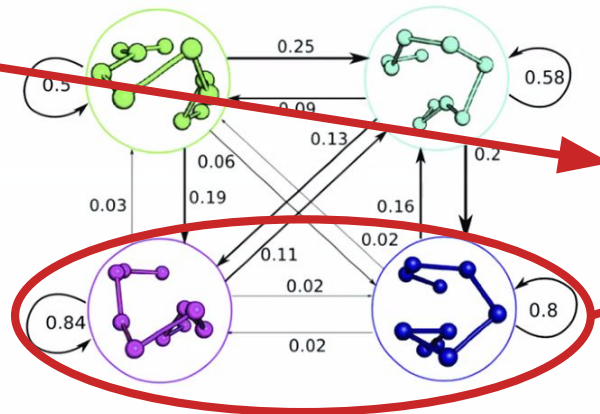


discrete conformational states



transition probabilities between states for lag time $\tau \rightarrow$ transition matrix $\mathbf{P}(\mathbf{t})$

➡ Model kinetics (folding pathways, equilibria, ...) - impossible from short MD trajectories

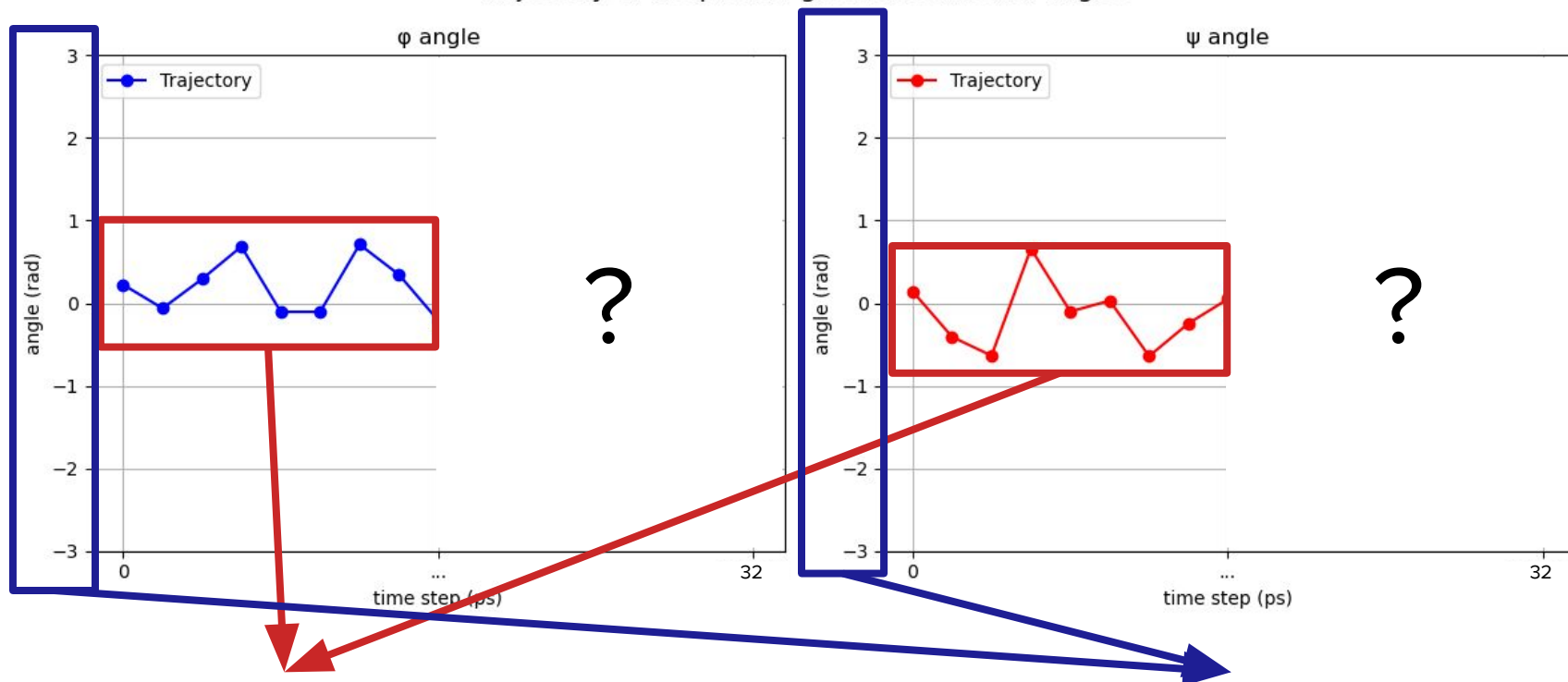


- Limited conformation possibilities
- One state as context

slide from the journal club : Week4_Dana_Alexa_VAMPNets_background

Introduction - Goal

Trajectory of the protein given the dihedral angles

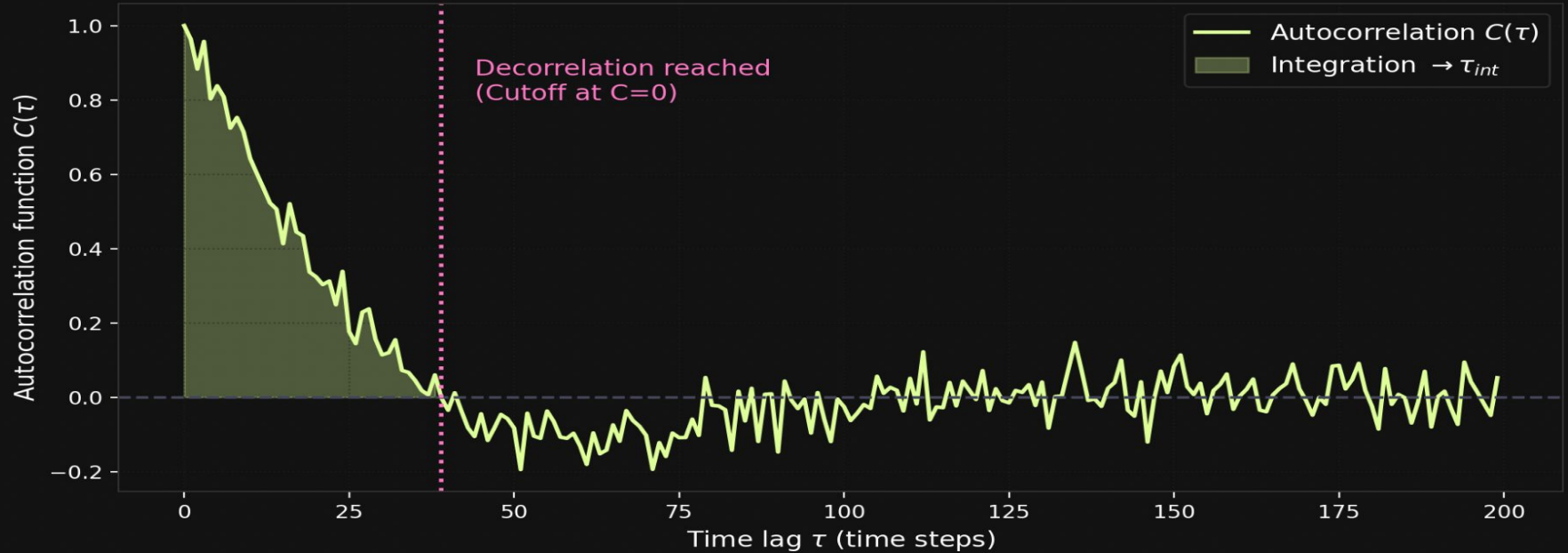


Context with more than one step

Non-discrete (continuous) states

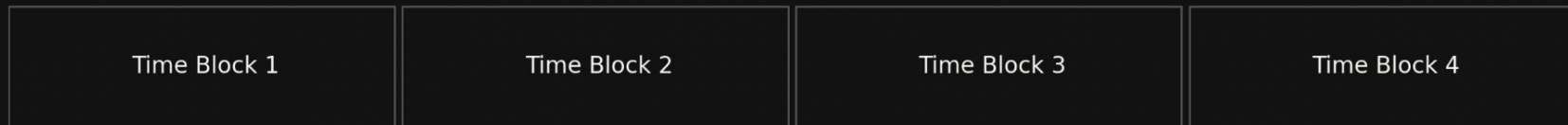
Data pipeline

Temporal Decorrelation Analysis via FFT

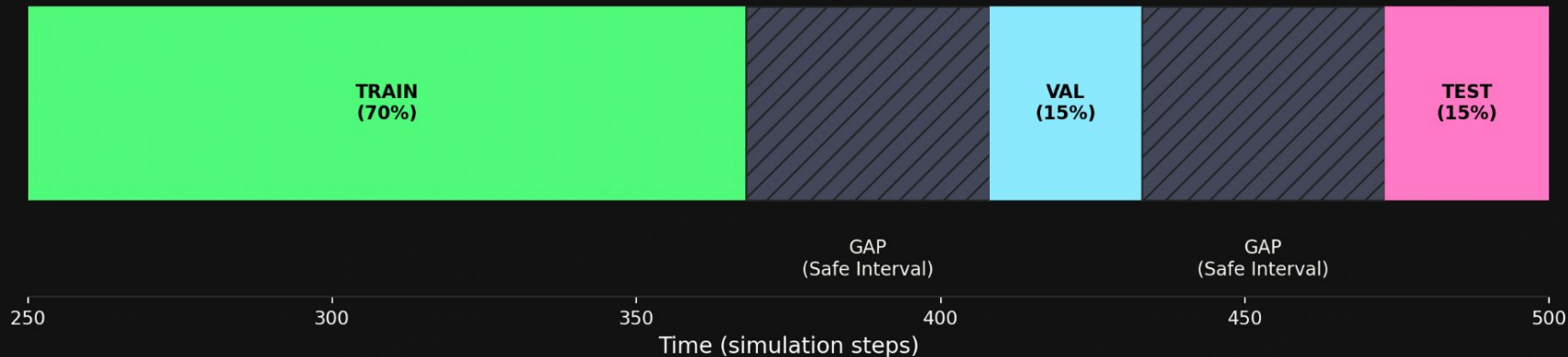


Data pipeline

Macro View: Uniform Temporal Coverage via Interleaved Blocks

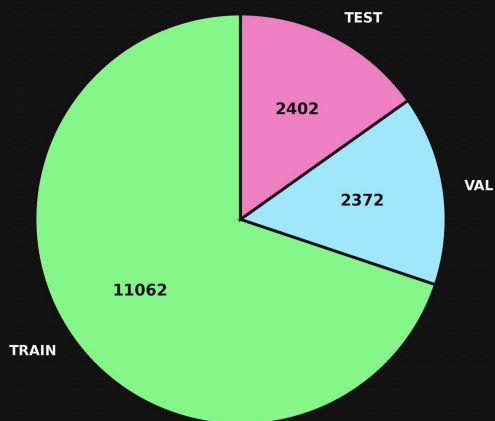


Micro View (Single Block): Hermetic Separation

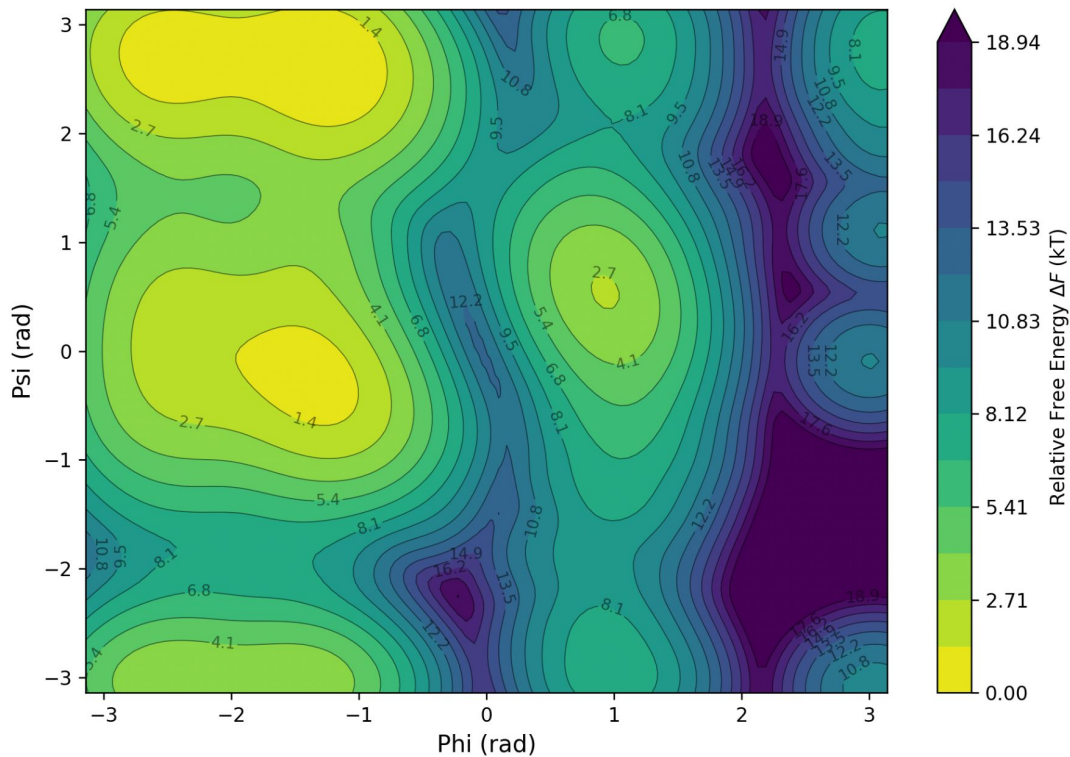


Dataset

Répartition des Tenseurs du Dataset



Free Energy Landscape of Alanine Dipeptide



Methodology

1. TRAINING DURATION

- **Total Budget: 150,000 Training Steps**

Total gradient updates performed on the dataset.

2. CONDITIONAL FLOW MATCHING OBJECTIVE

$$\mathcal{L}_{FM} = \mathbb{E}_{t, x_0, x_1} [\|v_{\theta}(x_t, t) - u_t(x_t|x_0, x_1)\|^2]$$

Minimizes MSE between predicted velocity field (v_{θ}) and target conditional vector field (u_t).

3. OPTIMIZATION: JENSEN-SHANNON DIVERGENCE (JSD)

Ensures generated dynamics match the underlying molecular physical distribution:

$$JSD(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M) \quad \text{where} \quad M = \frac{1}{2}(P + Q)$$

4. DATA AUGMENTATION

- Time Warp & Periodic Jitter
- Temporal Masking & Reflection

Enhances model robustness by simulating variations in molecular dynamics.

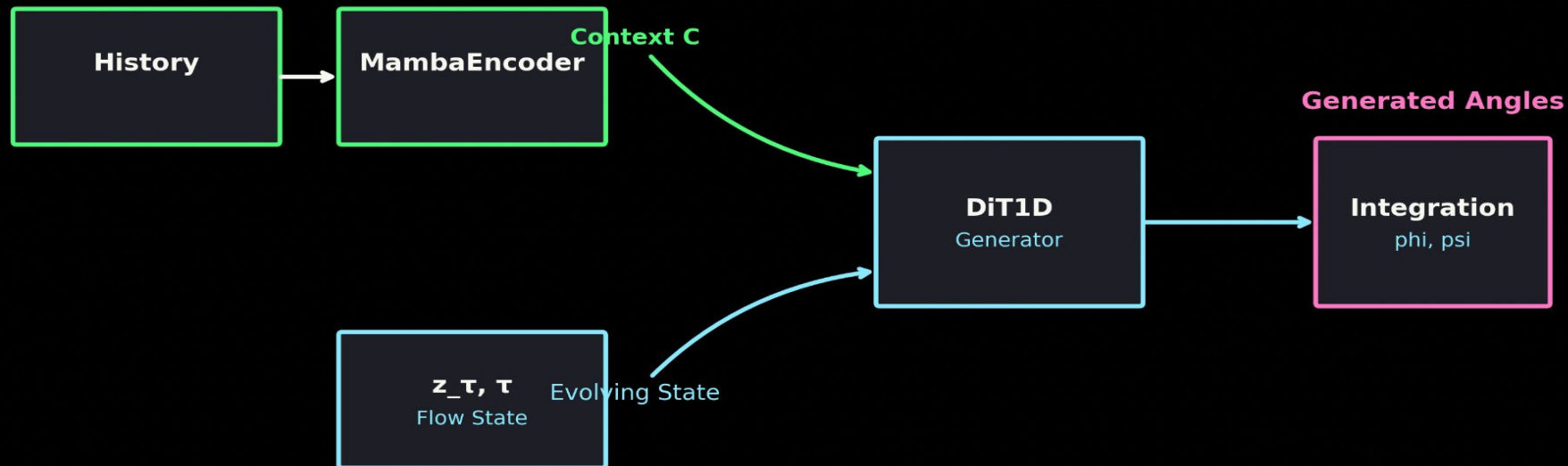
5. VALIDATION & RECOVERY

- Validation: Deterministic (shuffle=False)
- Recovery: Checkpoint loading via `resume_checkpoint`

Ensures reproducibility and fault tolerance.

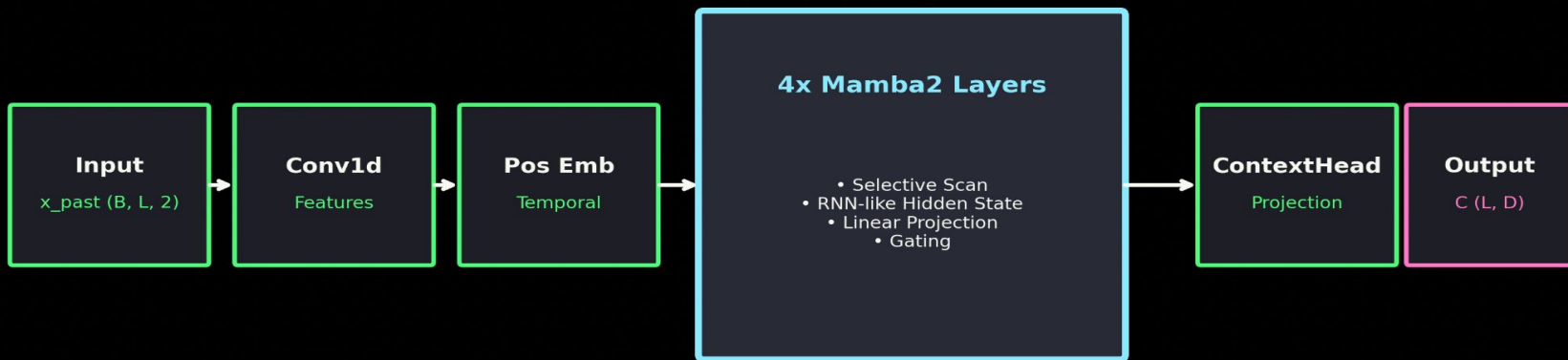
Model - Overall

Conditional Flow Molecule: Data Flow



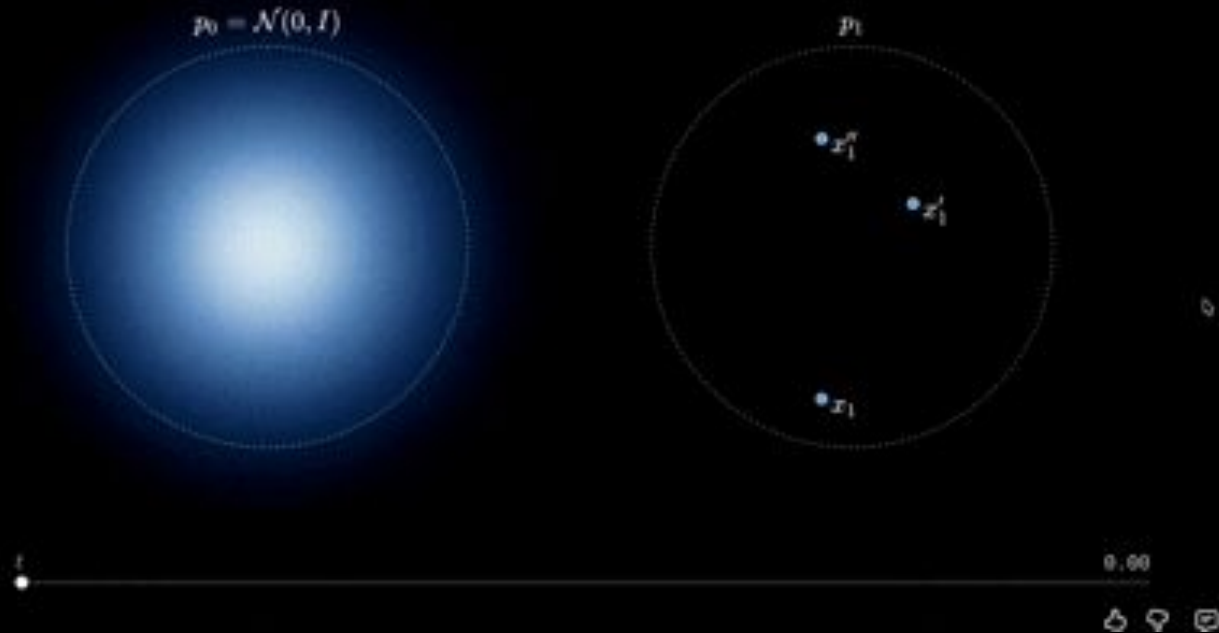
Model - Mamba

Mamba Encoder: Native Sequential Processing



Model - Flow Matching

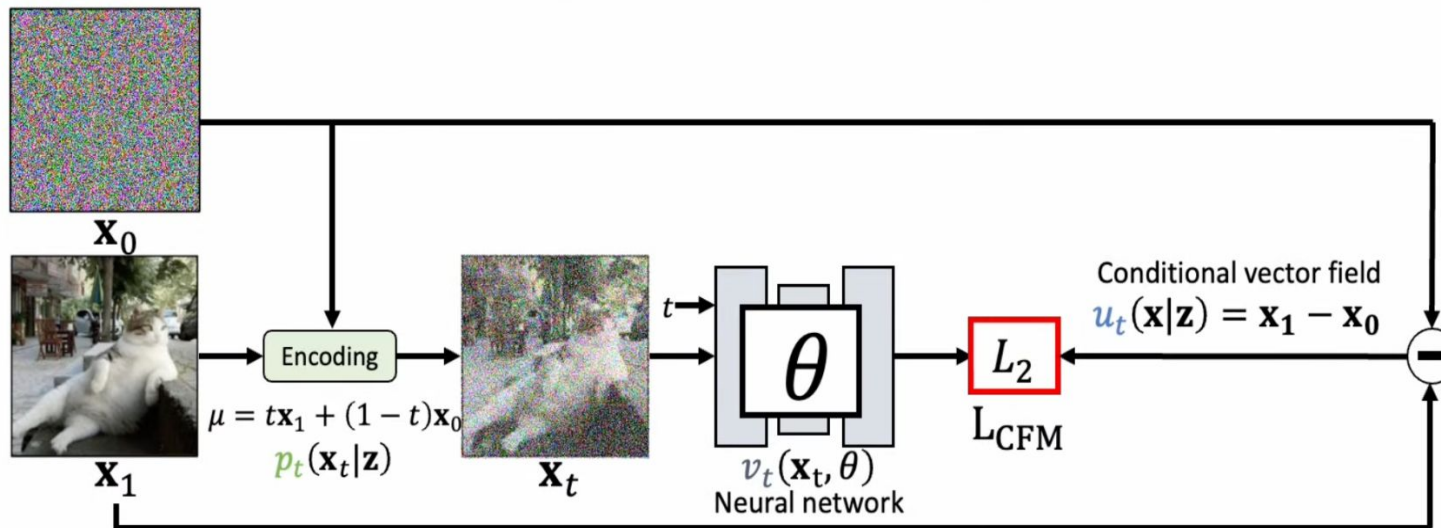
$$\frac{\partial p_t}{\partial t} = -\nabla \cdot (p_t v_t)$$



source: <https://www.youtube.com/watch?v=3mFNpeJQjmw>

Model - Flow Matching

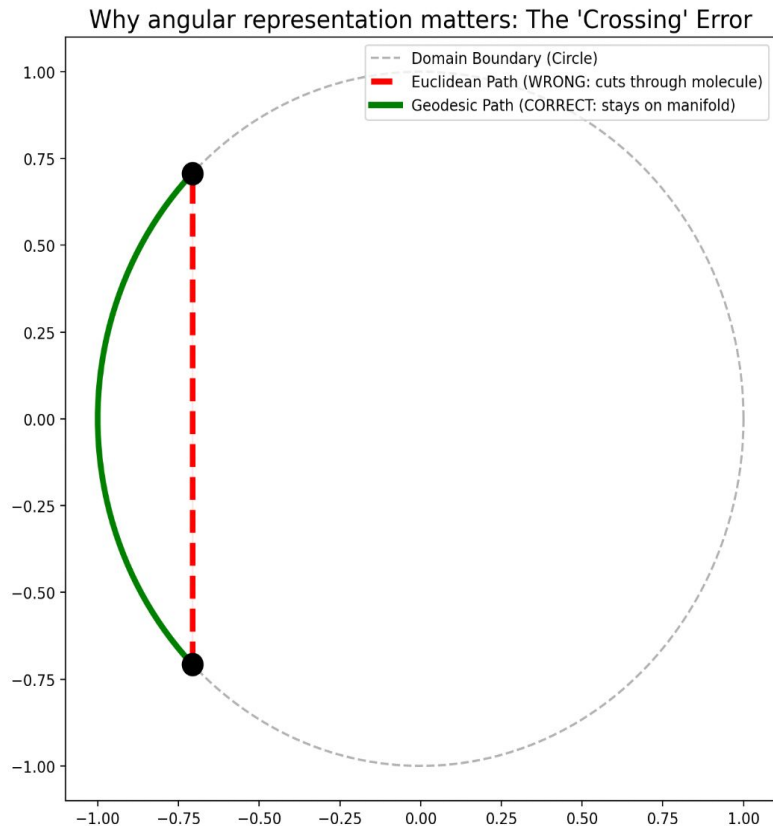
Flow matching



source:

https://www.youtube.com/watch?v=DDq_plfHqLs

Model - Diffusion



$$\Rightarrow \mathbf{u}_t(\mathbf{x}_1|\mathbf{x}_0) = (\mathbf{x}_1 - \mathbf{x}_0 + \pi) \pmod{2\pi} - \pi$$

$$\Rightarrow \mathbf{x}_t = (\mathbf{x}_0 + t \cdot \mathbf{u}_t(\mathbf{x}_1|\mathbf{x}_0) + \pi) \pmod{2\pi} - \pi$$

$$\Rightarrow \Phi(\theta) = \begin{pmatrix} \sin(\theta) \\ \cos(\theta) \end{pmatrix} \in \mathbb{R}^2$$

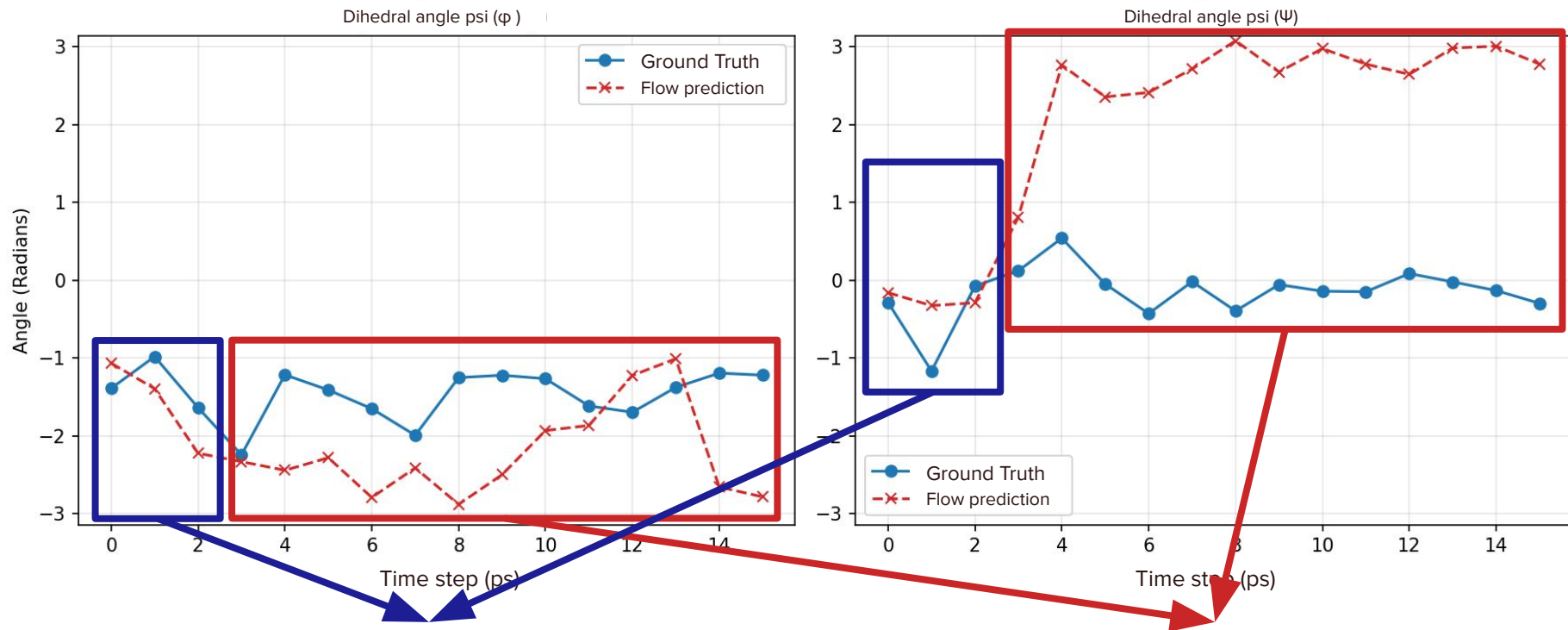
Map circular topology:

- Continuous manifold
- Stable gradient flow for the neural network.

Result - generating trajectory

Circular MAE : 29.86°

Flow matching evaluation



Non-stochastic process : used in the test-MAE MAE not relevant : stochastic process

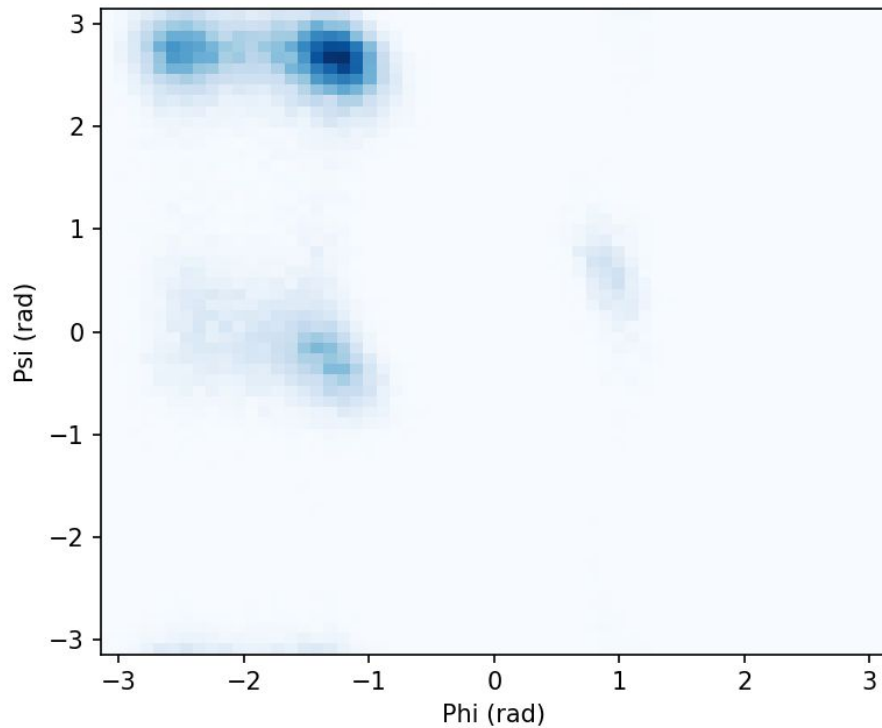
Result - comparing the distributions

Ramachandran Plot

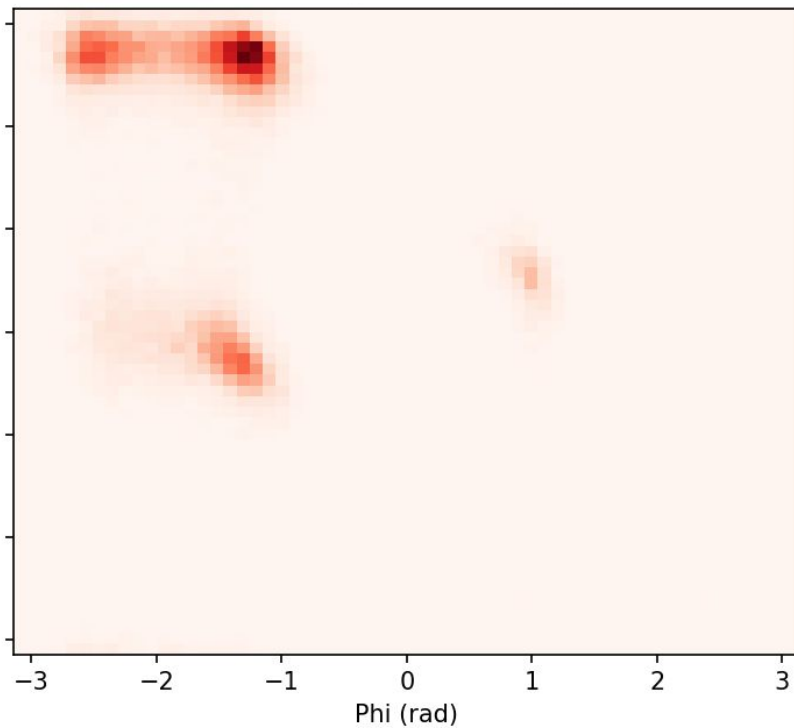
JSD Divergence

0.1678

Ground Truth



Generated



Result - comparing the distributions, JDS

JSD Divergence : 0.1678

$$M = 0.5 * (P + Q)$$

Mean distribution

$$D(P||M) = \sum_i P(i) \log \frac{P(i)}{M(i)}$$

$$D(Q||M) = \sum_i Q(i) \log \frac{Q(i)}{M(i)}$$

Dkl for each
distribution
compared to M

$$JSD(P||Q) = \frac{1}{2} (D(P||M) + D(Q||M))$$

Mean of the Dkl

JSD $\longrightarrow$ 0 $\Rightarrow P \sim Q$

JSD $\longrightarrow$ 1 $\Rightarrow P \neq Q$

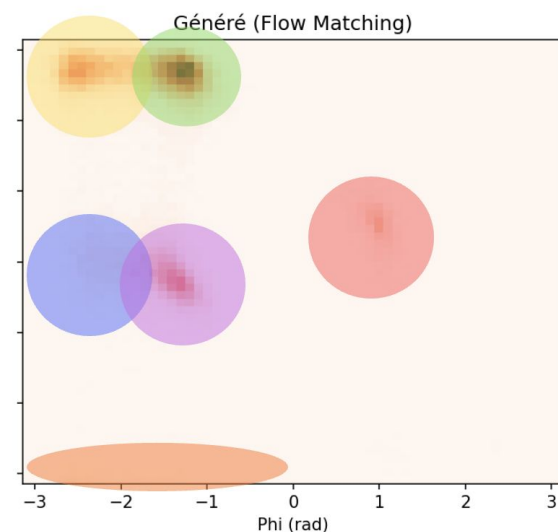
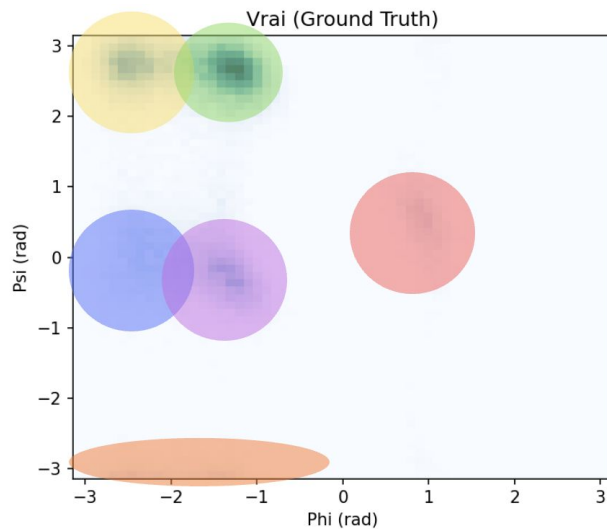
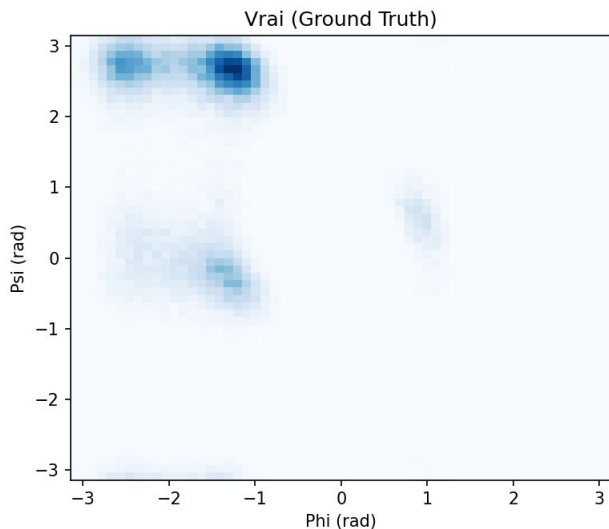
Result - comparing the distributions, K-Means

K-means
on the test

Cluster on GT
and generated

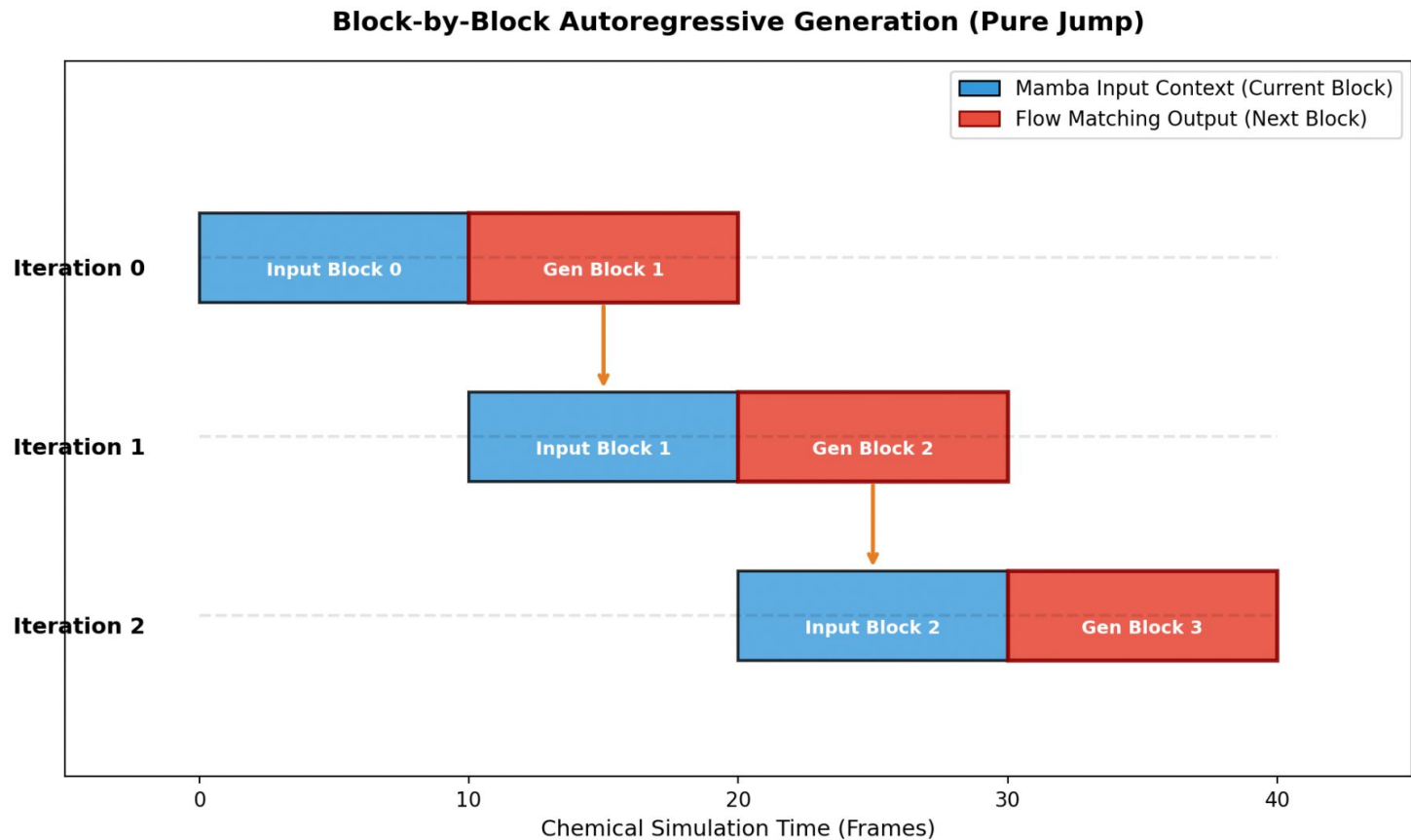
Transition matrix
for both graph

MAE to compare
both matrices

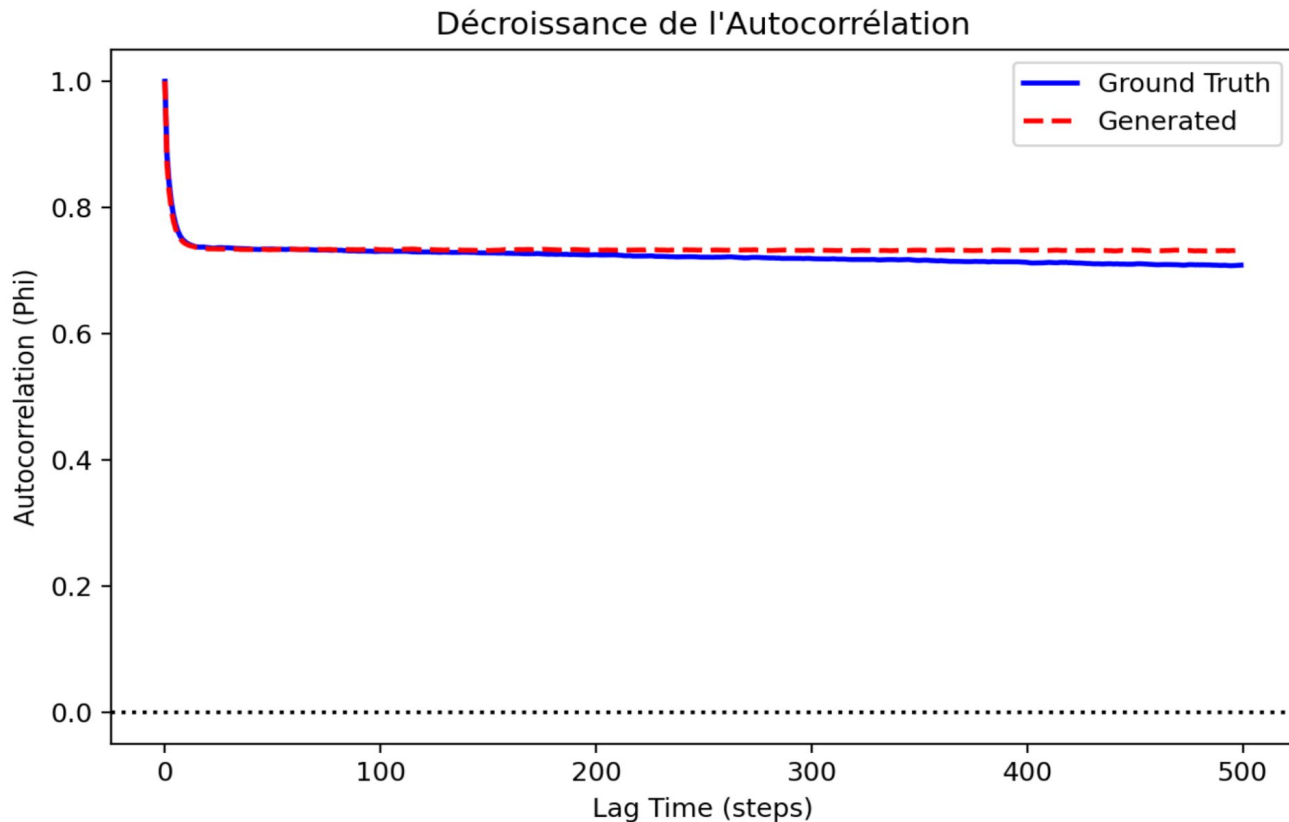


MAE : 0.015

Result – Rolling process

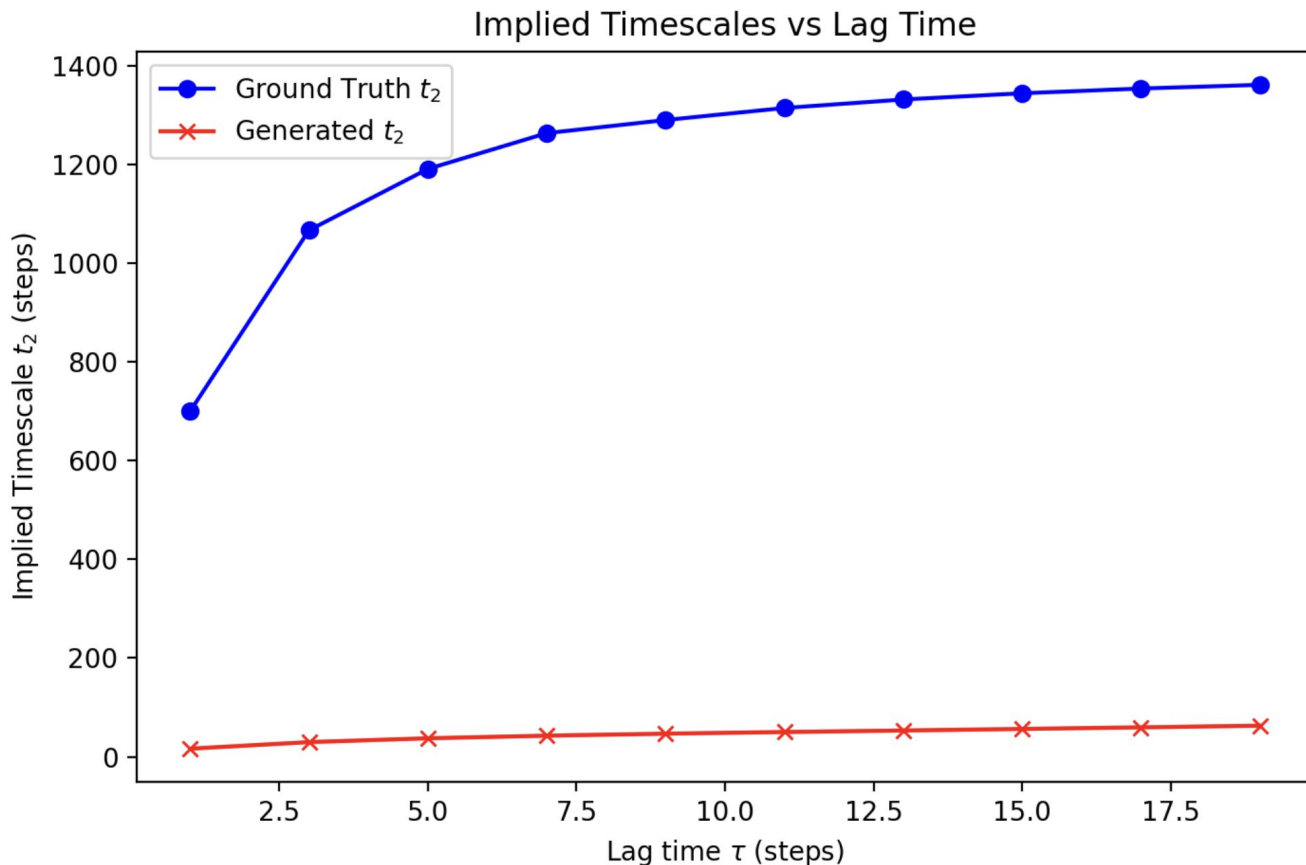


Result – Decorrelation analysis



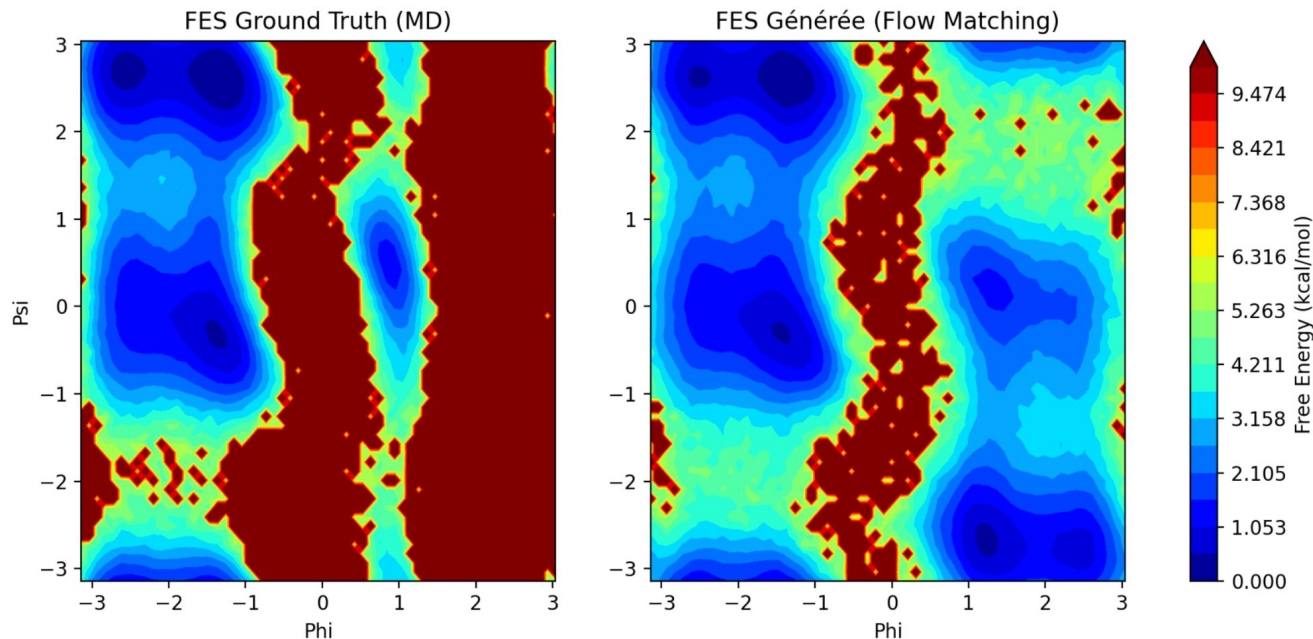
- **High Short-term Fidelity:** Excellent overlap at low lag times confirms the model accurately captures rapid, localized molecular dynamics.
- **Global Structural Consistency:** The curves do not collapse to zero, proving the model successfully retains long-term structural characteristics rather than reverting to random noise.

Result – Timescales analysis



While thermodynamics tells us *where* the molecule likes to go, timescales tell us *how fast* it gets there. Our model captures the correct shapes, but it accelerates the kinetics, crossing energy barriers roughly 13 times faster than real physics dictates.

Result – Energy barrier



$$F(\phi, \psi) = -k_B T \ln P(\phi, \psi)$$

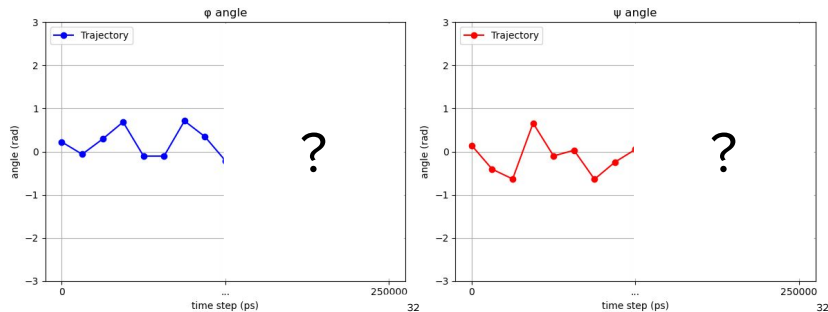
Barrier Flattening: The central energy barrier ($\phi \approx 0$) is heavily underestimated compared to the Ground Truth.

Unphysical Sampling: The model generates artificial low-energy states in the right quadrant ($\phi \in [1, 3]$) which are forbidden in real molecular dynamics.

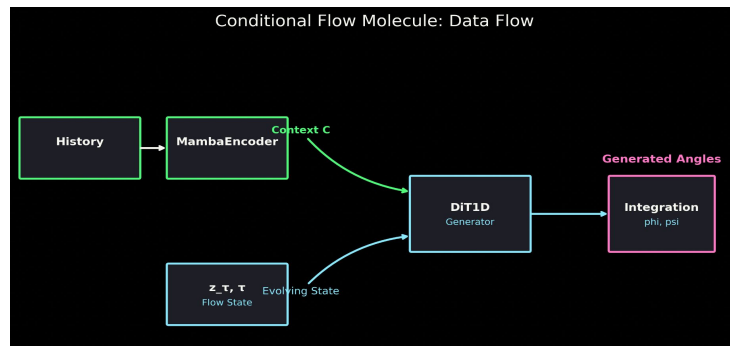
Kinetic Consequence: This explicit flattening of energy barriers visually explains why the generated implied timescales (t_2) collapse.

Conclusion

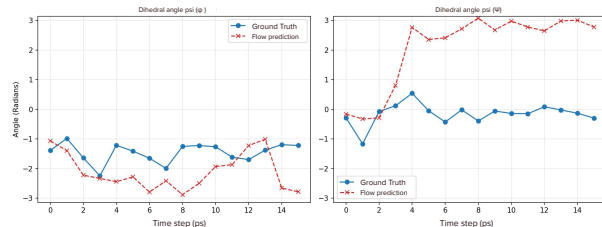
Trajectory of the protein given the dihedral angles



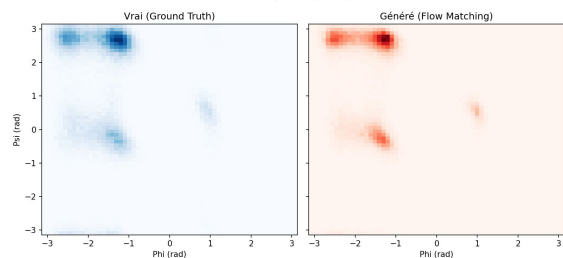
Conditional Flow Molecule: Data Flow



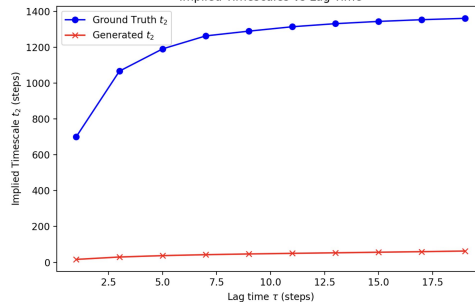
Flow matching evaluation



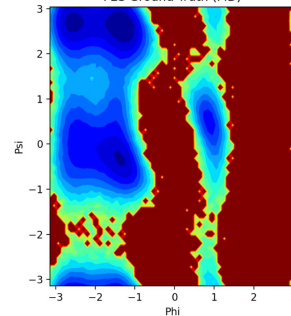
Ramachandran Plot | Divergence JSD : 0.1689



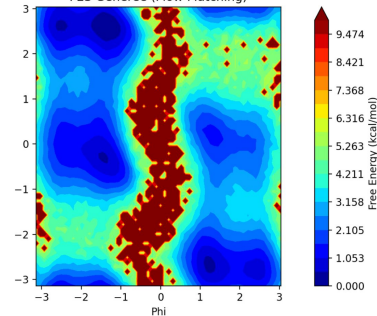
Implied Timescales vs Lag Time



FES Ground Truth (MD)



FES Générée (Flow Matching)



References

- [1] Ricky TQ Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. *Advances in neural information processing systems*, 31, 2018.
- [2] Mathias Schreiner, Ole Winther, and Simon Olsson. Implicit transfer operator learning: Multiple time-resolution models for molecular dynamics. *Advances in Neural Information Processing Systems*, 36:36449–36462, 2023.
- [3] Yuning Shen, Lihao Wang, Huizhuo Yuan, Yan Wang, Bangji Yang, and Quanquan Gu. Simultaneous modeling of protein conformation and dynamics via autoregression. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- [4] Hao Wu, Andreas Mardt, Luca Pasquali, and Frank Noé. Deep generative markov state models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 3979–3988, 2018.

Thank you for your attention
Thank you for this week